Problemset #2

1. Sure Tool Company is expected to pay a dividend of \$2 in the upcoming year. The risk-free rate of return is 4% and the expected return on the market portfolio is 14%. Analysts expect the price of Sure Tool Company shares to be \$22 a year from now. The beta of Sure Tool Company's stock is 1.25.

   What is the intrinsic value of Sure's stock today?

   k = .04 + 1.25 (.14 - .04); k = .165; .165 = (22 - P + 2)/P; .165P = 24 - P; 1.165P = 24 ; P = 20.60.

2. If Sure's intrinsic value is \$21.00 today, what must be its growth rate?

   k = .04 + 1.25 (.14 - .04); k = .165; .165 = 2/21 + g; g = .07.

3. Midwest Airline is expected to pay a dividend of \$7 in the coming year. Dividends are expected to grow at the rate of 15% per year. The risk-free rate of return is 6% and the expected return on the market portfolio is 14%. The stock of Midwest Airline has a beta of 3.00. The return you should require on the stock is

   6% + 3(14% - 6%) = 30%.

4. A company paid a dividend last year of \$1.75. The expected *ROE* for next year is 14.5%. An appropriate required return on the stock is 10%. If the firm has a plowback ratio of 75%, the dividend in the coming year should be

   g = .145 × .75 = 10.875%; \$1.75(1.10875) = \$1.94.

5. Old Quartz Gold Mining Company is expected to pay a dividend of \$8 in the coming year. Dividends are expected to decline at the rate of 2% per year. The risk-free rate of return is 6% and the expected return on the market portfolio is 14%. The stock of Old Quartz Gold Mining Company has a beta of -0.25. The intrinsic value of the stock is

   k = 6% + [-0.25(14% - 6%)] = 4%; P = 8/[.04 - (-.02)] = \$133.33.

6. Risk Metrics Company is expected to pay a dividend of \$3.50 in the coming year. Dividends are expected to grow at a rate of 10% per year. The risk-free rate of return is 5% and the expected return on the market portfolio is 13%. The stock is trading in the market today at a price of \$90.00.

   What is the approximate beta of Risk Metrics's stock?

   k = 13.9%; 13.9 = 5% + b(13% – 5%) = 1.11.

7. The market capitalization rate on the stock of Flexsteel Company is 12%. The expected *ROE* is 13% and the expected *EPS* are \$3.60. If the firm's plowback ratio is 75%, the P/E ratio will be

   g = 13% × 0.75 = 9.75%; .25/(.12 – .0975) = 11.11.

8. JCPenney Company is expected to pay a dividend in year 1 of \$1.65, a dividend in year 2 of \$1.97, and a dividend in year 3 of \$2.54. After year 3, dividends are expected to grow at the rate of 8% per year. An appropriate required return for the stock is 11%. The stock should be worth ______ today.

   Calculations are shown in the table below.

Yr	Dividend	PV of Dividend @ 11%
1	\$1.65	\$1.65/(1.11) = \$1.4865
2	\$1.97	\$1.97/(1.11)^2 = \$1.5989
3	\$2.54	\$2.54/(1.11)^3 = \$1.8572
	Sum	\$4.94

   P_3 = \$2.54 (1.08)/(.11 – .08) = \$91.44; *PV* of P_3 = \$91.44/(1.11)^3 = \$66.86; P_0 = \$4.94 + \$66.86 = \$71.80.

9. Assume that Bolton Company will pay a \$2.00 dividend per share next year, an increase from the current dividend of \$1.50 per share that was just paid. After that, the dividend is expected to increase at a constant rate of 5%. If you require a 12% return on the stock, the value of the stock is

   P_1 = 2 (1.05)/(.12 – .05) = \$30.00; *PV* of P_1 = \$30/1.12 = \$26.78; *PV* of D_1 = 2/1.12 = 1.79; P_0 = \$26.78 + \$1.79 = \$28.57.

10. The growth in dividends of Music Doctors, Inc. is expected to be 8% per year for the next two years, followed by a growth rate of 4% per year for three years; after this five-year period, the growth in dividends is expected to be 3% per year, indefinitely. The required rate of return on Music Doctors, Inc. is 11%. Last year's dividends per share were $2.75. What should the stock sell for today?

Calculations are shown below

Yr.	Dividend	PV of Dividend @ 11%
1	$\$2.75(1.08)$	$= \$2.97/(1.11) = \$2.6757$
2	$\$2.75(1.08)^2$	$= \$3.21/(1.11)^2 = \$2.6034$
3	$\$2.75(1.08)^2(1.04)$	$= \$3.34/(1.11)^3 = \$2.4392$
4	$\$2.75(1.08)^2(1.04)^2$	$= \$3.47/(1.11)^4 = \$2.2854$
5	$\$2.75(1.08)^2(1.04)^3$	$= \$3.61/(1.11)^5 = \$2.1412$
	Sum	$12.1449

$P_5 = 3.7164/(.11 - .03) = \$46.4544$; PV of $P_5 = \$46.4544/(1.11)^5 = \$27.5684$;
$P_0 = \$12.1449 + \$27.5684 = \$39.71$.

11. You purchase one September 50 put contract for a put premium of $2. What is the maximum profit that you could gain from this strategy?

-$200 + $5,000 = $4,800 (if the stock falls to zero).

12. The following price quotations were taken from the *Wall Street Journal*.

Stock Price	Strike Price	February
91 7/8	85	7 3/8
91 7/8	90	3 1/8
91 7/8	95	5/8

The premium on one February 90 call contract is

3 1/8 = $3.125 × 100 = $312.50. Price quotations are per share; however, option contracts are standardized for 100 shares of the underlying stock; thus, the quoted premiums must be multiplied by 100.

13. Suppose you purchase one WFM May 100 call contract at $5 and write one WFM May 105 call contract at $2.

The maximum potential profit of your strategy is _______ if both options are exercised.

-$100 - $5 = -$105; + $2 + $105 = $107; $2 × 100 = $200.

14. Suppose you purchase one WFM May 100 call contract at \$5 and write one WFM May 105 call contract at \$2.

    If, at expiration, the price of a share of WFM stock is \$103, your profit would be

    \$103 - \$100 = \$3 - (\$5 - \$2) = 0; \$0 × 100 = \$0.

15. Suppose you purchase one WFM May 100 call contract at \$5 and write one WFM May 105 call contract at \$2.

    The maximum loss you could suffer from your strategy is

    -\$5 + \$2 = -\$3 × 100 = -\$300.

16. Suppose you purchase one WFM May 100 call contract at \$5 and write one WFM May 105 call contract at \$2.

    What is the lowest stock price at which you can break even?

    x = \$100 + \$5 - \$2; x = \$103.

17. Top Flight Stock currently sells for \$53. A one-year call option with strike price of \$58 sells for \$10, and the risk-free interest rate is 5.5%. What is the price of a one-year put with strike price of \$58?

    $P = 10 - 53 + 58/(1.055)$; $P = 11.97$

18. Consider a one-year maturity call option and a one-year put option on the same stock, both with striking price \$45. If the risk-free rate is 4%, the stock price is \$48, and the put sells for \$1.50, what should be the price of the call?

    $C = 48 - [45/(1.04)] + 1.50$; C = \$6.23.

19. You purchased one corn future contract at \$2.29 per bushel. What would be your profit (loss) at maturity if the corn spot price at that time were \$2.10 per bushel? Assume the contract size is 5,000 ounces and there are no transactions costs.

    (\$2.10 - \$2.29 = -\$0.19) × 5,000 = -\$950.

20. You sold one corn future contract at \$2.29 per bushel. What would be your profit (loss) at maturity if the corn spot price at that time were \$2.10 per bushel? Assume the contract size is 5,000 ounces and there are no transactions costs.

    (\$2.29 - \$2.10 = \$0.19)× 5,000 = \$950.

21. Given a stock index with a value of \$1,500, an anticipated dividend of \$62 and a risk-free rate of 5.75%, what should be the value of one futures contract on the index?

$F = 1{,}500 \times (1 + .0575) - 62$; $F = 1{,}524.25$.

22. Suppose that the risk-free rates in the United States and in the United Kingdom are 4% and 6%, respectively. The spot exchange rate between the dollar and the pound is \$1.60/BP. What should the futures price of the pound for a one-year contract be to prevent arbitrage opportunities, ignoring transactions costs?

\$1.60(1.04/1.06) = \$1.57/BP.

23. Consider the following:

Risk-free rate in the United States	0.04/year
Risk-free rate in Australia	0.03/year
Spot exchange rate	1.67 A\$/\$

If the market futures price is 1.69 A\$/\$, how could you arbitrage?

Borrow Australian dollars in Australia, convert them to dollars, lend the proceeds in the United States, and enter futures positions to purchase Australian dollars at the current futures price.

0.5988(1.04) - 0.5917(1.03) = 0.013301; when this relationship is positive; action a will result in arbitrage profits.

24. Consider the following:

Risk-free rate in the United States	0.04/year
Risk-free rate in Australia	0.03/year
Spot exchange rate	1.67 A\$/\$

Assume the current market futures price is 1.66 A\$/\$. You borrow 167,000 A\$ and convert the proceeds to U.S. dollars and invest them in the U.S. at the risk-free rate. You simultaneously enter a contract to purchase 170,340 A\$ at the current futures prices (maturity of 1 year). What would be your profit (loss)?

Profit of 630 A\$

[A\$167,000/1.67 × 1.04 × 1.66] - (A\$167,000 × 1.03) = A\$630.

25. You hold a $50 million portfolio of par value bonds with a coupon rate of 10% paid annually and 15 years to maturity. How many T-bond futures contracts do you need to hedge the portfolio against an unanticipated change in the interest rate of 0.18%? Assume the market interest rate is 10% and that T-bond futures contracts call for delivery of an 8% coupon (paid annually), 20-year maturity T-bond.

0.9864485 × $50 M = $49,322,425; $50,000,000 - $49,322,425 = $677,575 loss on bonds; $100.00 - $82.97 = $17.03 × 100 = $1,703 gain on futures; $677,575/$1,703 = 398 contracts short.

26. You are given the following information about a portfolio you are to manage. For the long-term you are bullish, but you think the market may fall over the next month.

Portfolio Value	$1 million
Portfolio's Beta	0.60
Current S&P500 Value	1400
Anticipated S&P500 Value	1200

If the anticipated market value materializes, what will be your expected loss on the portfolio?

The change would represent a drop of (1,200 - 1,400)/1,400 = 14.3% in the index. Given the portfolio's beta, your portfolio would be expected to lose 0.6 × 14.3% = 8.57%.

27. You are given the following information about a portfolio you are to manage. For the long-term you are bullish, but you think the market may fall over the next month.

Portfolio Value	$1 million
Portfolio's Beta	0.60
Current S&P500 Value	1400
Anticipated S&P500 Value	1200

How many contracts should you buy or sell to hedge your position? Allow fractions of contracts in your answer.

The number of contracts equals the hedge ratio = change in portfolio value/profit on one futures contract = $85,700/$50,000 = 1.714. You should sell the contract because as the market falls the value of the futures contract will rise and will offset the decline in the portfolio's value.

28. Explain how a firm that has issued $1 million of long-term bonds with a fixed 6% interest rate can convert its fixed-rate debt into floating-rate debt. Give two numerical examples that show the possible outcomes, one favorable and one unfavorable.

The firm can enter a swap arrangement, committing to pay .06 × $1 million = $60,000 in exchange for receiving payments equal to $1 million times the LIBOR rate. If the LIBOR rate is 5, the cash inflow would be $1 million × .05 = $50,000. The net cash flow would be -$10,000 in this case, which is unfavorable. If the LIBOR rate is 8%, the firm will have a cash inflow of $1 million × .08 = $80,000. The net cash flow in this case is $20,000, which is favorable.